

Permian Basin Stratigraphy and Tectonic History

Ma	System	Series	Delaware Basin Formations	CBP & NW Shelf Formations	Midland Basin Formations	Tectonic Phase					
252	Permian	Ochoan	Dewey Lake	Dewey Lake	Dewey Lake	final basin fill/termination ↑ Post Orogenic Loading Rapid Subsidence; Onset of HC Generation (~ 50 million yrs. duration) ↓ Penn. – Early Permian Foreland Deformation ← <i>Early WFMP Event</i> ← <i>Strawn Event</i> ← <i>Atokan Event</i> ← <i>Morrow Event</i> (~ 30 million yrs. duration)	PERMIAN BASIN PHASE	Active Margin			
			Rustler	Rustler	Rustler						
			Salado	Salado	Salado						
			Castile								
		Guadalupian	Delaware Mtn Group	Lamar	Capitan				Tansill		
				Bell Canyon					Yates	Yates	
				Cherry Canyon					Seven Rivers	Seven Rivers	
									Queen	Queen	
				Brushy Canyon					Grayburg	Grayburg	
		Leonardian	Bone Spring	Upper San Andres					San Andres		
				Cutoff Member / 1 st Carb / Avalon	Lower San Andres				Holt / Upr Leonard	Clearfork	Holt / Upper Leonard
				1 st Sand	Glorieta					Spraberry	Upper Middle
				2 nd Carb	Upper Clearfork						Jo Mill
				2 nd Sand	Middle Clearfork						Lwr. L. SPBY Sh
	3 rd Carb / Stray Sd			Tubb		Dean					
	Wolfcampian	Wolfcamp	Lower Clearfork		Wolfcamp	Wolfcamp A					
			Wichita / Abq			Wolfcamp B					
299	Pennsylvanian	Virgilian	Cisco	Cisco	Cisco	Wolfcamp C1					
		Missourian	Canyon	Canyon	Canyon	Wolfcamp C2					
		Desmoinesian	Strawn	Strawn	Strawn	Wolfcamp D					
		Atokan	Atoka	Atoka	Atoka						
		Morrowan	Morrow	Morrow (NW Shelf Only)	Morrow						
			Barnett	U. Miss Lm > U. Barnett	U. Miss Lm > U. Barnett						
323	Mississippian	Meramecian – Osagean	L. Miss Lm	L. Miss Lm	L. Miss Lm						
			Kinderhookian	Woodford	Woodford	Woodford					
				Upper							
359	Devonian		Middle								
			Lower	Devonian (Thirtyone)	Devonian (Thirtyone)	Devonian (Thirtyone)					
			U. Niagaran	Upper Silurian (Wristen)	Upper Silurian (Wristen)	Upper Silurian (Wristen)					
419	Silurian		L. Niagaran								
			Alexandrian	Fusselman	Fusselman	Fusselman					
			Cincinnatian	Montoya	Montoya	Montoya					
444	Ordovician	Mohawkian	Bromide	Bromide	Bromide						
			Tulip Crk.	Tulip Crk.	Tulip Crk.						
		Chazyan	McKee Sd.	McKee Sd.	McKee Sd.						
			McClish	McClish	McClish						
			Waddell Sd.	Waddell Sd.	Waddell Sd.						
			Oil Creek	Oil Creek	Oil Creek						
		Canadian	Connell Sd.	Connell Sd.	Connell Sd.						
			Joins	Joins	Joins						
	Ozarkian	Ellenburger	Ellenburger	Ellenburger							
485	Cambrian	Upper	Bliss	Absent	Wilberns						
					Bliss / Hickory / Riley						
541	Precambrian		Precambrian Basement	Precambrian Basement	Precambrian Basement	Basement assembly and breakup of Rodinia supercontinent ← <i>Grenville Orogeny (1.3 to 1.0 bya)</i>	TOBOSA BASIN PHASE	Passive Margin			